Voltage: energy per charge

antocciphysics | TEACHER KEY | CPA

Learning target: Interpret potential difference as energy transferred per unit charge.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 1; 3 course days later.*
Repulsion: each force points away from the other object.

02. Explain a worked example (worked)
Higher potential by 3 V means potential-energy increase 6 J for +2 C.

03. Look and explain (model)
6 V supplies twice as much energy per coulomb.

04. Build the solution (practice)
12 J.

05. Use the idea (reason)
9 V means 9 J/C; total charge capacity and total stored energy require additional information.

06. Choose a method (mixed) *Review: Unit 6, Day 10; 12 course days later.*
Correct speed=2×60/0.6=200 m/s. The student used half the traveled path and gets half the speed.

07. Exit check (exit)
4 J per coulomb.

08. STRETCH: Apply it another way (stretch)
6 and 18 J; equal voltage does not imply equal total energy.

Voltage: energy per charge

antocciphysics | TEACHER KEY | HONORS

Learning target: Interpret potential difference as energy transferred per unit charge.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 1; 3 course days later.*
Polarization redistributes charge. Nearer opposite charge can attract more strongly than farther like charge repels; net charge may remain zero.

02. Explain a worked example (worked)
Higher potential by 3 V means potential-energy increase 6 J for +2 C.

03. Represent and interpret (model)
6 V supplies twice as much energy per coulomb.

04. Solve and justify (practice)
12 V; 15 J.

05. Analyze the reasoning (reason)
9 V means 9 J/C; total charge capacity and total stored energy require additional information.

06. Choose a method (mixed) *Review: Unit 6, Day 10; 12 course days later.*
True speed 340 m/s; incorrect speed≈272 m/s, biased low.

07. Exit check (exit)
6 J and 12 J; 18 J total.

08. STRETCH: Apply it another way (stretch)
ΔU=-100 microjoules; field work +100 microjoules. Negative charge loses potential energy moving to higher potential.